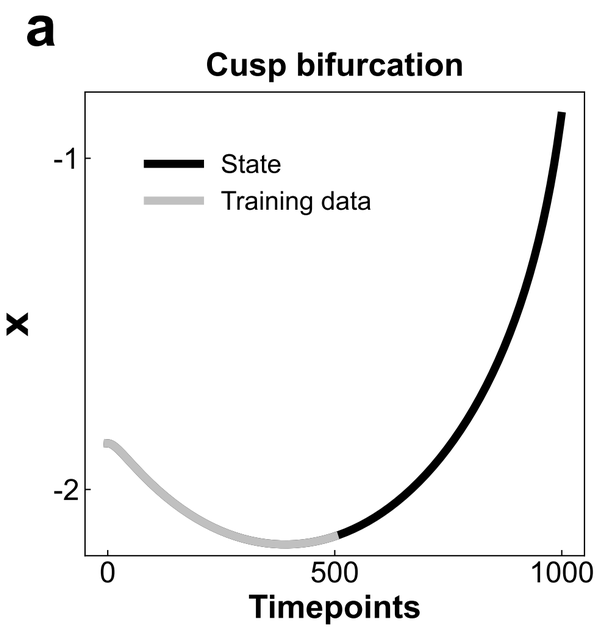
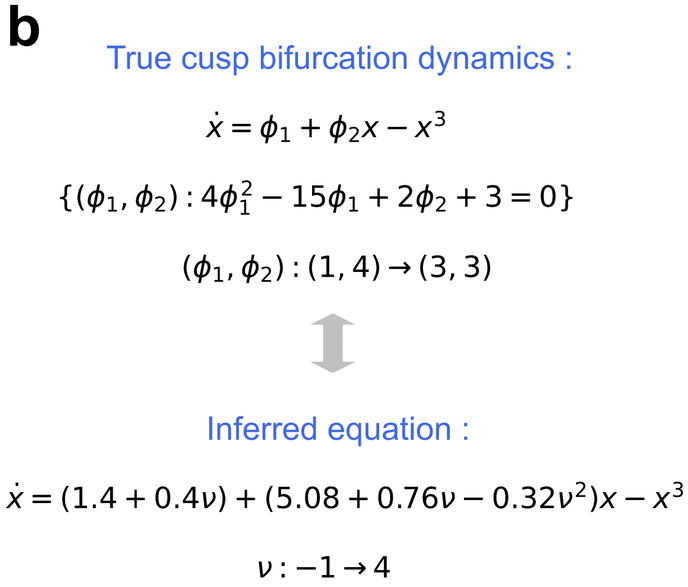
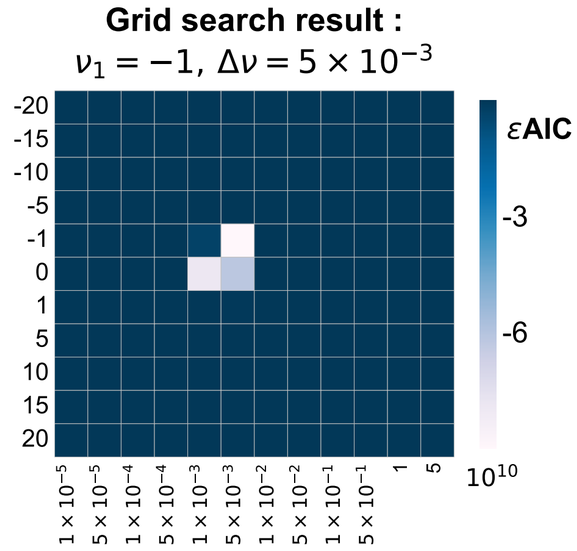
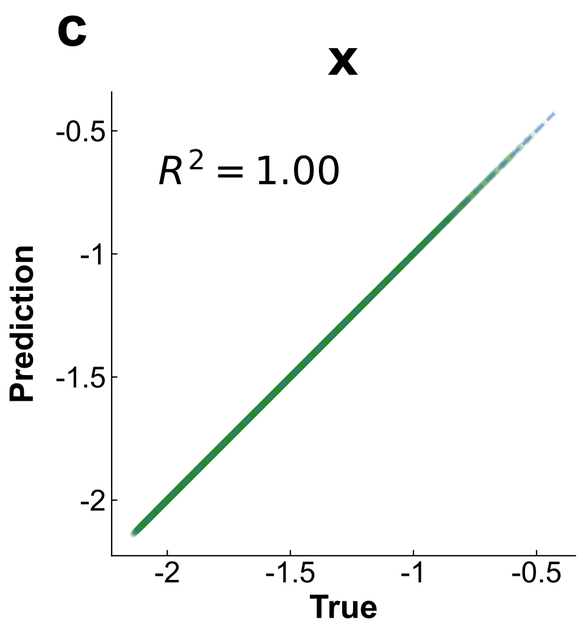

a
Cusp bifurcation
State
Training data
x
-1
-2
0
500
1000
Timepoints


b
True cusp bifurcation dynamics :
$\dot{x} = \phi_1 + \phi_2 x - x^3$
$\{(\phi_1, \phi_2) : 4\phi_1^2 - 15\phi_1 + 2\phi_2 + 3 = 0\}$
$(\phi_1, \phi_2) : (1, 4) \rightarrow (3, 3)$
Inferred equation :
$\dot{x} = (1.4 + 0.4\nu) + (5.08 + 0.76\nu - 0.32\nu^2)x - x^3$
$\nu : -1 \rightarrow 4$


Grid search result :
$\nu_1 = -1,\ \Delta\nu = 5 \times 10^{-3}$
-20
-15
-10
-5
-1
0
1
5
10
15
20
1×10^{-5}
5×10^{-5}
1×10^{-4}
5×10^{-4}
1×10^{-3}
5×10^{-3}
1×10^{-2}
5×10^{-2}
1×10^{-1}
5×10^{-1}
1
5
εAIC
-3
-6
10^{10}


c
x
$R^2 = 1.00$
Prediction
-0.5
-1
-1.5
-2
True
-2
-1.5
-1
-0.5